Project MO V1
Physical Fit-Test Plan

Eight-coupon mechanical validation program for the V1 enclosure

Document	Project MO V1 Physical Fit-Test Plan
Revision	1.0
Date	14 September 2026
Owner	Heitor de Moraes Fagundes
Scope	Fastener, switch, limb, USB-C, microphone, ESP32, amplifier and speaker coupons
Status	Prepared for supplier quotation and physical testing

Important: Keep the editable CAD and exported 3MF files as the source package. Do not issue G-code until the selected supplier confirms the exact printer, nozzle, material profile and slicing workflow.

Document purpose

This plan provides a repeatable method for checking whether the V1 enclosure interfaces fit their real components before the full enclosure is manufactured. It separates printing defects from CAD clearance problems and creates evidence for controlled revisions.

1. Test controls

1.1 General procedure

• Test every coupon before changing the production enclosure.

• Use the real production component and its intended orientation.

• Keep electronics disconnected and unpowered during mechanical fitting.

• Apply only gentle hand pressure. Stop if a part, tab, boss or printed wall begins to bend or crack.

• Perform three insertion/removal cycles for ordinary sliding fits. Use only one removal cycle for fragile snap tabs and self-tapping threads.

• Photograph each fitted coupon from the front and side, and record the printed coupon ID.

• Do not sand, drill, file or enlarge a fit surface before the baseline result is recorded.

1.2 Result classifications

Result	Definition	Action
PASS	Secure fit, normal insertion and removal, correct alignment.	Accept the tested interface.
TIGHT	Part fits but requires excessive force or visibly stresses the print.	Review clearance; test a controlled increase.
LOOSE	Part fits but rattles, rotates, shifts or is not retained.	Review clearance or retaining geometry.
FAIL	Part cannot be inserted, retained, aligned or fastened as intended.	Stop; measure and revise CAD.
PRINT DEFECT	Warping, blobs, layer separation or missing features make the result unreliable.	Reprint before changing CAD.

1.3 Tools and materials

• All eight PETG coupons and the matching physical components

• Digital calipers (recommended resolution: 0.01 mm)

• Manual Phillips screwdriver for M2 and M3 self-tapping screws

• Known-good USB cables for connector-access checks

• Camera or phone, permanent coupon IDs and this printed record

Use a manual screwdriver only for the coupon tests. Do not use a powered driver. If print cleanup requires a sharp tool, ask an experienced adult to handle it and record the unmodified condition first.

2. Pre-test print inspection

Complete this inspection before inserting any component. A failed print must not be used to justify a CAD change.

Check	Requirement	Result
Identity	Label and coupon ID match the intended component.	[ ] Pass [ ] Issue
Material	Supplier confirms the requested PETG or reports any substitution.	[ ] Pass [ ] Issue
Base	Base is flat, stable and free of visible warping.	[ ] Pass [ ] Issue
Layers	No cracks, major separation or missing sections.	[ ] Pass [ ] Issue
Fit surfaces	No blobs, collapsed walls or loose strands on critical surfaces.	[ ] Pass [ ] Issue
Openings	Holes and slots are fully open and retain their intended shape.	[ ] Pass [ ] Issue
Thin features	Guides, clips, bosses and circular extensions are intact.	[ ] Pass [ ] Issue
Engraving	Identification is readable; cosmetic imperfections are acceptable.	[ ] Pass [ ] Issue
Dimensions	Critical dimensions are reasonably consistent with the reference table.	[ ] Pass [ ] Issue

Inspection record

Coupon ID	Material	Layer height	Nozzle	Inspection result

3. Coupon reference matrix

Dimensions below are the coupon design references. Measurements from a printed part may differ slightly because of the printer and material process.

Coupon	Critical geometry	Primary validation
Fastener	M2 boss Ø7 x 4.5; pilot Ø1.6 x 4.5; clearance Ø2.2. M3 boss Ø8 x 10; pilot Ø2.6 x 8; clearance Ø3.4.	Thread formation, boss integrity and clearance holes.
Switch	45 x 45 x 3 base; centered Ø20.0 opening.	Panel fit and snap-tab engagement.
Limb	LP 5 x 5 x 3; LS 5.6 x 5.6 through; AP 6 x 6 x 3; AS 6.4 x 6.4 through.	Leg and arm peg/socket clearances.
USB-C	Opening 9.60 x 3.80, R1.80; two 3 mm walls; 4.50 edge offset; 13.8 internal guide spacing; 8 mm guide depth; 0.50 chamfer.	Dedicated USB-C interface and cable access.
Microphone	3 mm face; 8 mm slot, R4 end; cradle Ø14.60 inner / Ø17.60 outer; 4 mm height; 3 x 3 tab, 1 mm height.	Microphone retention and acoustic-opening alignment.
ESP32	PCB 51.0 x 27.5 x 1.6; clear 51.4 x 27.9; Ø4 x 5.5 supports; support spacing 41 x 14; 10.4 guide projection; USB opening 10 x 5, R1.	Horizontal board mount and its own USB opening.
Amplifier	Clear area 18.4 x 17.4; three-sided guide 3 mm thick x 3 mm high; open insertion side.	MAX98357A board fit in C-shaped guide.
Speaker	Ø45 outer / Ø40.6 inner; full ring 10 mm; two additional 8 mm sections; side chords 19.0 outer / 16.5 inner.	40 mm speaker retention and cradle flexibility.

The dedicated USB-C coupon and the ESP32 USB opening are different interfaces. Do not substitute the 10 x 5 mm ESP32 opening for the 9.60 x 3.80 mm dedicated USB-C opening.

4. Fastener coupon

Validate the production M2 and M3 screw interfaces before using them in the enclosure.

Reference geometry

Feature	Nominal value
M2 pilot boss	Ø7 x 4.5 mm; Ø1.6 x 4.5 mm blind pilot
M2 clearance hole	Ø2.2 mm through
M3 pilot boss	Ø8 x 10 mm; Ø2.6 x 8 mm blind pilot
M3 clearance hole	Ø3.4 mm through

Required parts

M2 x 8 self-tapping plastic screw; M3 x 10 self-tapping plastic screw; manual screwdriver.

Procedure

1. Inspect both bosses and confirm the pilot holes are clear.
2. Pass each screw through its clearance hole without turning it; it should not form a thread.
3. Start each screw vertically in its labelled pilot using slow manual turns.
4. Stop if torque rises sharply, the boss whitens, bends or cracks, or the screw wanders.
5. Tighten only until firmly seated. Remove once and reinstall once to assess serviceability.

Pass criteria

- Clearance screw passes by hand with little lateral movement.
- Pilot provides moderate, consistent resistance and the screw does not spin freely.
- Boss remains intact and the screw can be reinstalled with firm retention.

Record

Record M2 and M3 independently: insertion behavior, approximate turns, visible damage, first tightening result and reinstall result.

Interface	Result	Observations / measured change
M2 clearance		
M2 pilot/boss		
M3 clearance		
M3 pilot/boss		

5. Switch coupon

Validate the round switch opening and snap retention in a 3 mm panel.

Reference geometry

Feature	Nominal value
Panel	45 x 45 x 3 mm
Opening	Ø20.0 mm centered

Required parts

intended 20 mm round switch.

Procedure

1. Confirm the opening is clear and inspect the switch retaining tabs.
2. Align the switch in its production orientation and press evenly using finger pressure.
3. Inspect the front flange and rear retaining tabs.
4. Gently check rotation and movement. Remove once to assess serviceability.

Pass criteria

• No tool or excessive force is needed.

• Front flange sits flat.

• Tabs engage behind the 3 mm plate and the switch does not rattle or rotate.

• Coupon and switch tabs remain undamaged.

Record

Record insertion force qualitatively, flange seating, tab engagement, rotation, movement and removal condition.

6. Limb-interface coupon

Validate both leg and arm peg/socket tolerance pairs.

Reference geometry

Feature	Nominal value
LP	5.0 x 5.0 x 3.0 mm leg peg
LS	5.6 x 5.6 mm through leg socket
AP	6.0 x 6.0 x 3.0 mm arm peg
AS	6.4 x 6.4 mm through arm socket

Required parts

production leg; production arm; matching peg/socket references as available.

Procedure

1. Identify LP, LS, AP and AS and inspect their corners.
2. Insert each matching interface straight, without twisting it into place.
3. Check fully seated depth, lateral play and resistance to gentle withdrawal.
4. Perform three controlled insertion/removal cycles unless damage appears.

Pass criteria

- Each interface reaches its intended seated position.
- No cracking, severe whitening or permanent deformation.
- Fit is secure but remains removable by hand; no obvious wobble.

Record

Record LP, LS, AP and AS separately. Note whether the peg, socket or both would require revision.

7. Dedicated USB-C coupon

Validate the separate USB-C interface, perpendicular-wall relationship and cable access.

Reference geometry

Feature	Nominal value
Opening	9.60 x 3.80 mm, R1.80
Opening offset	Nearest edge 4.50 mm from adjacent wall
Walls	3.00 mm
Guide	13.8 mm internal spacing; 18.8 x 1.5 mm rails; 8 mm depth
Entrance	0.50 mm equal-distance chamfer

Required parts

dedicated USB-C module/interface; known-good USB-C cable.

Procedure

1. Insert the module into the U-shaped guide in its production orientation.
2. Confirm that it reaches the intended stop and aligns with the perpendicular opening.
3. With power disconnected, insert the cable through the opening into the connector.
4. Check plug seating, shell clearance and cable removal without levering the module.
5. Remove and reinsert the module three times if its connector and guide remain undamaged.

Pass criteria

• Guide accepts and locates the module without excessive force or rattle.
• Connector is centered and accessible through the opening.
• Cable seats fully without touching or lifting the surrounding wall.
• Chamfer eases insertion and shows no damage.

Record

Record module fit, alignment in both axes, cable seating, wall contact and any required opening or guide change.

8. Microphone coupon

Validate the repurposed ventilation slot as the acoustic opening and the circular microphone cradle.

Reference geometry

Feature	Nominal value
Face plate	3.00 mm
Acoustic slot	8.00 mm wide with R4 end; locally extended through coupon edge
Cradle	Ø14.60 inner; 1.50 wall; Ø17.60 outer; 4.00 mm height
Tab	3 x 3 mm; 1.00 mm height
Alignment	Cradle center shares a slot center point

Required parts

production microphone module.

Procedure

1. Inspect the slot, cradle opening and retaining tab.

2. Orient the microphone so its acoustic port faces the slot and its pins/cable have clearance.
3. Press the module into the cradle using even pressure on its rigid edges, not on sensitive components.
4. Check center alignment through the slot and gentle movement in all directions.
5. Remove and reinstall up to three times if the module and tab remain undamaged.

Pass criteria

• Module seats without damaging the cradle or electronics.

• Acoustic port aligns with the slot and remains unobstructed.

• Cradle and tab retain the module without rattle while allowing service removal.

Record

Record insertion direction, retention, acoustic-port alignment, pin/cable clearance and any cradle-diameter change.

9. ESP32 coupon

Validate the horizontal ESP32 mount beneath the display together with the ESP32-specific USB opening.

Reference geometry

Feature	Nominal value
PCB reference	51.0 x 27.5 x 1.6 mm
Clear area	51.40 x 27.90 mm
Supports	Four Ø4 x 5.5 mm; centers spaced 41.0 x 14.0 mm
Corner guides	L profiles; 10.40 mm projection; overall references 55 x 32 mm
USB opening	10.0 x 5.0 mm, R1.0, in adjacent 3 mm wall

Required parts

production ESP32 board; known-good USB cable matching the ESP32 connector.

Procedure

1. Confirm all four supports and four corner guides are intact.
2. Orient the board horizontally as installed beneath the display.
3. Insert it evenly so the PCB rests against all four support faces.
4. Check that the guides contact the board edges without bending the PCB.
5. With power disconnected, insert the USB cable through the 10 x 5 opening and confirm full seating.
6. Remove and reinstall the board three times if no damage or excessive stress appears.

Pass criteria

- PCB seats on all four supports and remains flat.
- No guide forces the PCB to bow; no significant lateral or vertical rattle.
- USB connector aligns with the opening and accepts the cable without wall contact.
- Board remains removable for service.

Record

Record seating at each support, PCB movement, guide stress, USB alignment, cable access and any directional clearance change.

> Do not confuse this 10 x 5 mm ESP32 USB opening with the dedicated USB-C coupon opening of 9.60 x 3.80 mm.

10. Amplifier coupon

Validate the MAX98357A amplifier board in the three-sided guide.

Reference geometry

Feature	Nominal value
Clear area	18.40 x 17.40 mm
Guide	3.00 mm thick x 3.00 mm high
Insertion	Left side open; top, bottom and right sides guided

Required parts

production MAX98357A amplifier board.

Procedure

1. Inspect the C-shaped guide and confirm the open insertion side is clear.
2. Orient the board as installed and slide it through the open side.
3. Seat it to the right-side stop without forcing components against the guide.
4. Check movement, pin clearance and removal access.
5. Perform three sliding cycles if the PCB and guide remain undamaged.

Pass criteria

• Board reaches its intended position without excessive force.

• Guide does not overlap or contact sensitive board components.

• Board is located without significant rattle and remains removable.

Record

Record sliding resistance, final seating, movement, component/pin clearance and proposed directional change.

11. Speaker coupon

Validate the 40 mm speaker diameter, retention depth and flexible side openings.

Reference geometry

Feature	Nominal value
Cradle diameter	Ø45.00 outer / Ø40.60 inner; 2.20 mm radial wall
First section	Full circular ring, 10.00 mm
Extensions	Upper and lower sections, additional 8.00 mm; total 18.00 mm
Side openings	19.00 mm outer chord / 16.50 mm inner chord

Required parts

production 40 mm speaker.

Procedure

1. Inspect the ring, both extended sections and both side openings.
2. Hold the speaker by its rigid frame; do not press the diaphragm.
3. Align it with the cradle and insert gradually while observing cradle flex.
4. Confirm the speaker reaches its intended depth and sits evenly.

5. Check gentle movement and removal without twisting or prying against the frame.

Pass criteria

- Speaker frame enters without excessive force or permanent cradle deformation.
- Speaker seats evenly and is retained without noticeable rattle.
- Side openings flex without cracking and return to their original position.
- Diaphragm, terminals and wires remain unobstructed.

Record

Record insertion force, seated depth, retention, cradle recovery, terminal clearance and any diameter or chord change.

12. Master results record

Use one row for each printed coupon. If a coupon is reprinted, add a new row and preserve the original result.

Coupon / ID	Print inspection	Fit result	Cycles	Evidence	CAD action
Fastener					
Switch					
Limb interface					
USB-C					
Microphone					
ESP32					
Amplifier					
Speaker					

Detailed observation sheet

Field	Entry
Coupon / print ID	
Test date and tester	
Printer / supplier	
Material / color	
Layer height / nozzle	
Measured dimensions	
Insertion orientation	
Cycle 1	
Cycle 2	
Cycle 3	
Final classification	
Photo filenames	
Recommended action	

13. CAD revision rules

Apply these rules after the physical evidence is complete.

Observation	Check first	Controlled response
Tight in one direction	Blobs, seam placement, elephant foot and component orientation.	Change only the affected directional clearance; begin with a small controlled adjustment.
Tight everywhere	Material shrinkage, printer calibration and nominal component size.	Compare printed and CAD dimensions before increasing the full profile.
Loose	Correct component variant and measurement method.	Reduce only the locating clearance or revise the intended retention feature.
Misaligned opening	Component fully seated and correctly oriented.	Move the opening or mount reference; do not enlarge automatically.
Cracked thin feature	Print orientation, layer bonding and supplier settings.	Distinguish process failure from insufficient feature thickness.
One bad copy	Compare with another print if available.	Treat as a print defect until repeatability is established.

Revision workflow

1. Preserve the original result, photographs and baseline dimensions.
2. Identify whether the cause is the component, print process or CAD geometry.
3. Change one functional dimension at a time. A typical first trial is 0.10-0.20 mm, but the measured evidence takes priority.
4. Save a new Fusion version and record the exact feature and reason for the change.
5. Export a newly named 3MF and reprint only the affected coupon.
6. Repeat the same test procedure and compare results.
7. Transfer a successful coupon revision to the production model only after confirmation.

Known deferred production action: apply the verified 0.50 mm guide-entry chamfer to the editable Main Body USB-C guide after the physical USB-C fit result confirms it. Regenerate the production Main Body 3MF only after that controlled update.

14. Supplier print record and sign-off

The supplier should slice the 3MF models for the confirmed machine. The settings below are requirements or reporting fields, not pre-generated G-code.

Item	Requested / confirmed value
Source files	Eight individual Project MO V1 coupon 3MF files
Material	PETG; supplier to state manufacturer and product
Layer height	Target 0.20 mm unless supplier recommends and documents another value
Infill	Target 20% unless supplier recommends and documents another value
Supports	Designed for no supports in the reviewed orientations; supplier must report any exception
Nozzle	Supplier to confirm
Printer	Supplier to confirm
Orientation	Supplier to preserve fit-critical surfaces and document changes
G-code	Generated by supplier only after machine/profile confirmation

Final authorization	Entry
All eight tests complete	[ ] Yes [ ] No
Failed interfaces re-tested	[ ] Yes [ ] No [ ] Not applicable
Production CAD updated	[ ] Yes [ ] No [ ] Not required
Production interference check	[ ] Pass [ ] Fail [ ] Pending
Approved by	
Date	
Notes	

End of document